Derivation of the Disturbance Equations

From the compressible Navier–Stokes equations to the four disturbance ODEs solved by `pyMack`

Personal reference note

July 1, 2026

Abstract

This note derives, step by step, the ordinary differential equations that govern an infinitesimal disturbance riding on a compressible laminar boundary layer. Starting from the compressible Navier–Stokes equations, we split every field into a steady mean plus a small disturbance, drop products of disturbances to linearise, invoke the parallel-flow assumption, and insert the normal-mode ansatz $q' = \hat{q}(y)\, e^{\mathrm{i}(\alpha x - \omega t)}$. The result is four coupled ODEs in the wall-normal coordinate y — continuity (with the equation of state folded in), x- and y-momentum, and energy — exactly as assembled in `pymack/temporal_solver.py`. The mean-flow profiles that appear as coefficients are taken as given (they are computed in the companion *Mean Boundary Layer* note); the discretisation of these ODEs into matrices and the eigenvalue solve are carried out in the companion *Numerical Methods* note. A closing section explains what the resulting eigenmodes physically *are* — first and second (Mack) modes — and the four idealisations behind the equations.

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How to read this. Boxes marked **“What problem is solved here”** state exactly which equation and unknown are involved at each stage. *In plain words* notes give the intuition; grey Remark asides hold secondary detail you can skip on a first read.

Notation. Overbars are mean (base-flow) quantities; hats are complex disturbance amplitudes. The continuous wall-normal amplitude is $\hat{q}(y) = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$. A prime on a mean quantity is $\mathrm{D} \equiv \mathrm{d}/\mathrm{d}y$.

Base flow & transport		**Disturbance / eigenvalue**	
$\overline{U}, \overline{T}, \overline{\rho}{=}1/\overline{T}$	mean velocity, temp., density	$\hat{q}$	amplitude shape, Eq. (6)
$\overline{\mu}, \overline{\kappa}$	viscosity, conductivity	$\hat{u}, \hat{v}, \hat{T}, \hat{p}$	velocity/temp./pressure amplitudes
$M_e,\ Re,\ Pr,\ \gamma$	Mach, Reynolds, Prandtl, c_p/c_v	$\alpha,\ \omega$	wavenumber, frequency
$\nu_R{=}1/(Re\overline{\rho})$	viscous prefactor	$c{=}\omega/\alpha$	complex phase speed; c_r, c_i its parts
$\mathcal{C}_\kappa, \mathcal{C}_d, \mathcal{L}_c$	conduction/dissipation groups	M_{rel}	relative Mach number, §5

1 What stability theory asks, and the plan

A laminar boundary layer is a smooth, steady flow that eventually becomes turbulent. The first stage of that process is the amplification of tiny disturbances — noise, roughness, free-stream turbulence — always present in a real flow. *Linear* Stability Theory asks the cleanest version of the question:

> If we add one infinitesimally small wave to the laminar flow, does that wave **grow** or **decay** as it travels?

We will write that small wave as $q'(x, y, t)$ — shorthand for the collection of all disturbance quantities (velocity, temperature, pressure; defined precisely in Eq. (5)). "Infinitesimal" is the key word: if q' is small enough, products of small quantities are negligible and its governing equations become *linear*.

Why this becomes an eigenvalue problem. After linearising, the coefficients of the equations depend only on the wall-normal coordinate y. Because they are *constant in x and t*, substituting a single trial wave $q' \propto e^{\mathrm{i}(\alpha x - \omega t)}$ turns every x- and t-derivative into multiplication by a constant; the equations then no longer involve x or t at all, only the wave's wall-normal shape. The requirement that a *non-zero* shape exist is precisely an **eigenvalue condition** — solved numerically in the companion *Numerical Methods* note.

In plain words. *An eigenvalue problem is the statement "$\mathsf{A}\hat{q} = \lambda\hat{q}$": for most λ the only solution is $\hat{q} = 0$ (no wave), but for special values λ — the eigenvalues — a non-zero shape $\hat{q}$ (the eigenfunction) exists. Here the eigenvalue carries the growth rate, so its sign tells us grow or decay.*

LST is one stage of a longer story (Figure 2): disturbances first *enter* the layer (receptivity), then grow linearly (LST — this document), then break down nonlinearly to turbulence. We treat only the linear-growth stage; §6 states exactly what that leaves out.

The computation itself is a short pipeline (Figure 3):

1. compute the steady **mean (base) flow** (companion *Mean Boundary Layer* note);
2. **linearise** and insert a wave **ansatz** to get ordinary differential equations in y (§3–4, this note);
3. **discretise** those ODEs into matrices with Chebyshev collocation (companion *Numerical Methods* note);
4. solve the resulting **eigenvalue problem** for the growth rate (companion *Numerical Methods* note);
5. sweep parameters to draw **neutral curves** and integrate **N-factors** (companion *Numerical Methods* note).

This note delivers stage 2: it takes the mean flow of stage 1 as a known coefficient and produces the disturbance ODEs that stages 3–5 discretise and solve.

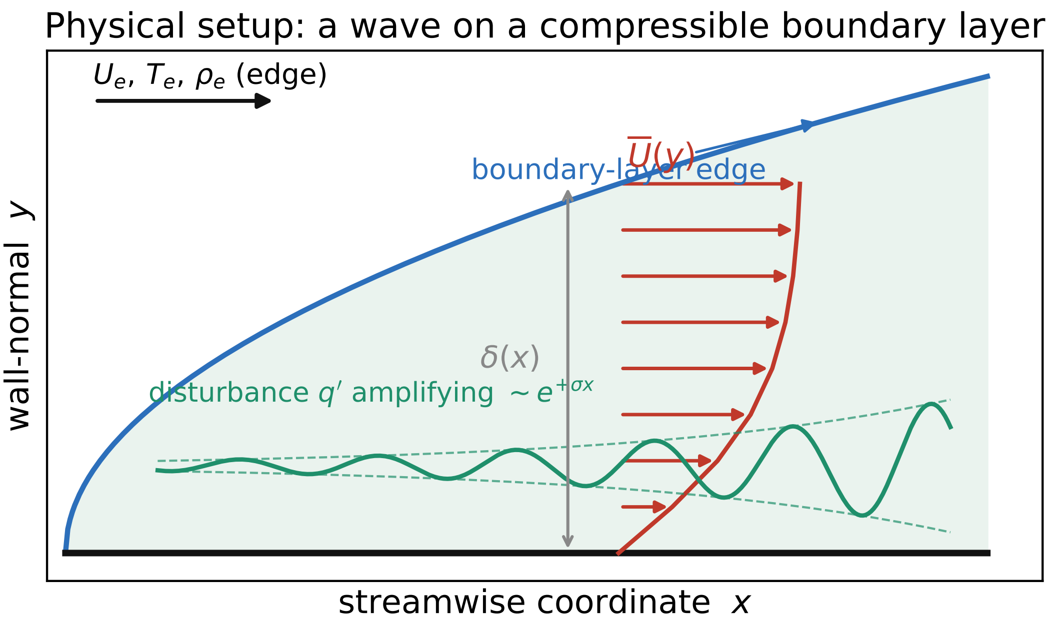

Figure 1: The physical setup. A steady compressible boundary layer grows along a wall; its mean velocity $\overline{U}(y)$ and temperature $\overline{T}(y)$ vary across the layer of thickness $\delta(x)$. LST tracks one small disturbance $q'(x, y, t)$ riding in this layer and asks whether it amplifies downstream.

2 Overview: from the governing equations to the matrix problem

Before the derivation itself, here is the whole arc in one picture — where these equations come from and where they lead.

From the governing equations to the matrix problem the code solves. This note, together with the companion *Numerical Methods* note, is a single chain of transformations carrying the *full governing equations* (top — nonlinear, unsolvable by hand) into the *one matrix problem the code actually solves* (bottom — boxed). Those two are the forms worth remembering; everything in between is how we get from the first to the last.

$$\partial_t\rho + \nabla\cdot(\rho\mathbf{u}) = 0, \qquad \rho\,\tfrac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + \nabla\cdot\boldsymbol{\tau},$$
$$\rho\,c_p\,\tfrac{\mathrm{D}T}{\mathrm{D}t} = \tfrac{\mathrm{D}p}{\mathrm{D}t} + \nabla\cdot(\kappa\nabla T) + \Phi, \qquad p = \rho R T$$

the **governing equations**: nonlinear, compressible Navier–Stokes in (x, y, t) for $\mathbf{q} = (u, v, T, p, \rho)$ — Eqs. (1)–(4)

↓ split $\mathbf{q} = \overline{\mathbf{q}} + \mathbf{q}'$, subtract the mean (it solves the equations on its own), drop products of the small primes (§3, Steps 1–2)

linear PDEs for the disturbance $\mathbf{q}' = (u', v', T', p', \rho')$ in (x, y, t)
every coefficient is now a *known* mean profile $\overline{U}(y), \overline{T}(y), \ldots$

↓ parallel-flow assumption, then the normal-mode ansatz $\mathbf{q}' = \hat{q}(y)\,e^{\mathrm{i}(\alpha x - \omega t)}$, so $\partial_x \to \mathrm{i}\alpha,\ \partial_t \to -\mathrm{i}\omega$ (§3, Steps 3–5)

$$\mathbf{L}(\alpha, c;\, y, \mathrm{D})\,\hat{q}(y) = \mathbf{0}$$

four coupled **ODEs in** y for the wall-normal shape $\hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})$ — Eqs. (8)–(11), the endpoint of this note

↓ Chebyshev collocation: $\mathrm{D} \to D_1,\ \mathrm{D}^2 \to D_2$, each mean coefficient → a diagonal matrix, the four fields stacked into one vector (companion *Numerical Methods* note)

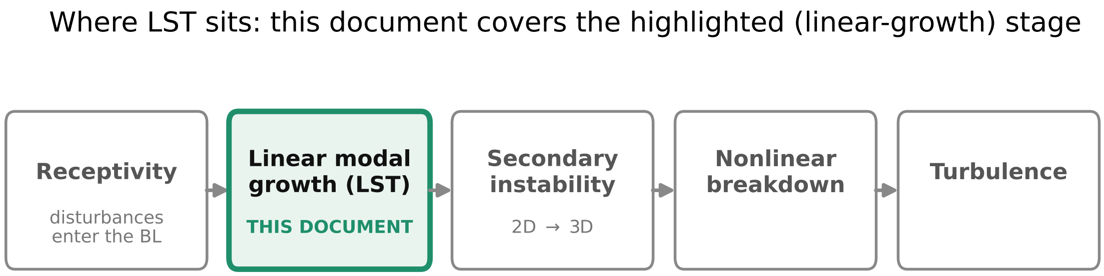

Figure 2: The transition process. This document covers the highlighted linear-modal-growth stage (which produces the e^N amplification); receptivity and nonlinear breakdown are separate problems (§6).

$$\boxed{A(\alpha)\,\boldsymbol{\phi} = c\,B(\alpha)\,\boldsymbol{\phi}} \qquad A, B \in \mathbb{C}^{4n\times 4n}$$

the **discrete generalised eigenvalue problem** `temporal_solver.py` assembles and solves — derived in the companion *Numerical Methods* note

The top line is a system of nonlinear partial differential equations no one can solve by hand; the bottom line is finite linear algebra. The eigenvalue carries the answer: `temporal_solver.py` fills the two matrices A, B and calls `scipy.linalg.eig`(A, B), which returns the complex phase speed c (the eigenvalue) and the grid-sampled mode shape $\boldsymbol{\phi}$ (the eigenvector). Holding α real and solving for c is the *temporal* problem; holding ω real and regrouping the same ODEs by powers of α turns the last line into the *spatial* quadratic problem. **The present note derives the two middle lines** — the linear PDEs and the four ODEs in y; the discretisation and eigenvalue-problem steps that produce the boxed bottom line are carried out in the companion *Numerical Methods* note.

Where the coefficients come from. Every overbarred symbol in the chain — $\overline{U}(y)$, $\overline{T}(y)$, $\overline{\mu}$, $\overline{\kappa}$ and their y-derivatives — is a *known* function once the steady laminar base flow has been solved. That boundary-value problem, its non-dimensionalisation, and the Mack length/Reynolds scales are the subject of the companion *Mean Boundary Layer* note; here we simply treat those profiles as given data.

3 Linearisation and the normal-mode ansatz

We now perturb the mean flow by an infinitesimal wave and reduce the governing PDEs to ordinary differential equations in y — the equations the rest of the LST pipeline discretises and solves.

The flow obeys the compressible Navier–Stokes equations — conservation of mass, momentum, and energy for a viscous, heat-conducting perfect gas:

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} + \rho\,\nabla\cdot\mathbf{u} = 0, \qquad \text{(mass)} \qquad (1)$$

$$\rho\frac{\mathrm{D}\mathbf{u}}{\mathrm{D}t} = -\nabla p + \nabla\cdot\boldsymbol{\tau}, \qquad \text{(momentum)} \qquad (2)$$

$$\rho\,c_p\frac{\mathrm{D}T}{\mathrm{D}t} = \frac{\mathrm{D}p}{\mathrm{D}t} + \nabla\cdot(\kappa\nabla T) + \Phi, \qquad \text{(energy)} \qquad (3)$$

$$p = \rho R T, \qquad \text{(state)} \qquad (4)$$

where $\mathbf{u} = (u, v)$ is velocity, ρ density, T temperature, p pressure, μ viscosity, κ conductivity, c_p specific heat, R the gas constant, $\mathrm{D}/\mathrm{D}t = \partial_t + \mathbf{u}\cdot\nabla$ the material derivative, $\boldsymbol{\tau}$ the Newtonian viscous

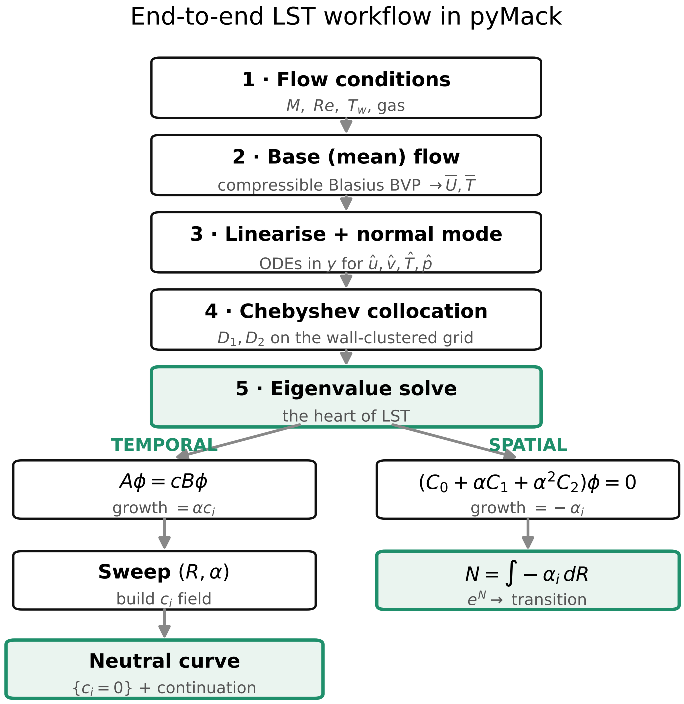

Figure 3: End-to-end workflow. The single eigenvalue solve (stage 5) forks into a *temporal* route (real wavenumber α in, complex phase speed c out; used for neutral curves) and a *spatial* route (real frequency ω in, complex α out; used for N-factors).

stress tensor, and $\Phi = \boldsymbol{\tau} : \nabla \mathbf{u}$ the dissipation function (both linear in velocity gradients). These are the same equations solved for the steady mean flow in the companion *Mean Boundary Layer* note; here we apply them a second time, linearised, for the disturbance. The reduction proceeds in five steps.

Step 1 — split into mean plus disturbance.

$$u = \overline{U}(y) + u',\ v = v',\ T = \overline{T}(y) + T',\ p = \overline{p} + p',\ \rho = \overline{\rho}(y) + \rho', \qquad |\cdot'| \ll |\bar{\cdot}|. \tag{5}$$

($\overline{v} = 0$ for a parallel flow.) The collection $q' = (u', v', T', p', \rho')$ is the disturbance.

Step 2 — drop products of disturbances. Substitute (5) into (1)–(4). The mean satisfies the equations by itself and cancels; any term with two primes is the product of two small numbers and is dropped. What remains is *linear* in q' with coefficients made of the known mean profiles.

Step 3 — parallel-flow assumption. The layer grows slowly, so locally $\overline{q} = \overline{q}(y)$ only; the coefficients depend on y alone.

Caveat. This neglects the $O(1/R)$ streamwise growth of the mean flow and the eigenfunction; the rigorous remedy is the Parabolised Stability Equations (PSE), which march the disturbance downstream retaining that slow x-variation (§6).

Step 4 — the normal-mode ansatz.

$$q'(x,y,t) = \hat{q}(y)\, e^{\mathrm{i}(\alpha x - \omega t)}, \qquad \hat{q} = (\hat{u}, \hat{v}, \hat{T}, \hat{p})^{\mathsf{T}}, \tag{6}$$

with α the streamwise wavenumber ($2\pi/\alpha$ = wavelength), ω the frequency, $c = \omega/\alpha$ the complex phase speed, $\hat{q}(y)$ the wall-normal shape.

2D wave. We take spanwise wavenumber $\beta = 0$. A general oblique wave $\propto e^{\mathrm{i}(\alpha x + \beta z - \omega t)}$ travels at angle $\psi = \arctan(\beta/\alpha)$; compressibility breaks Squire's theorem, so the *first* mode is most unstable oblique ($\psi \sim 55°$) while the *second* mode peaks at $\psi = 0$. The 2D solver thus captures the second mode faithfully but under-predicts the first (§6).

Substitution gives $\partial_x \mapsto \mathrm{i}\alpha$, $\partial_t \mapsto -\mathrm{i}\omega$, $\partial_y \mapsto \mathrm{D} \equiv \mathrm{d}/\mathrm{d}y$, collapsing the PDEs to coupled ODEs in y.

Step 5 — eliminate density. The linearised state law gives $\rho'/\overline{\rho} = \gamma M_e^2 \hat{p} - \hat{T}/\overline{T}$, removing ρ'. The state then has four fields $\hat{u}, \hat{v}, \hat{T}, \hat{p}$, with $\hat{p}$ the pressure amplitude scaled by $\rho_e U_e^2$ (so $\gamma M_e^2 \hat{p} = p'/p_e$).

A worked linearisation (continuity), so the mechanics are explicit. Take mass conservation (1), $\partial_t \rho + \nabla\cdot(\rho \mathbf{u}) = 0$. Insert (5); the steady mean satisfies $\partial_x(\overline{\rho}\overline{U}) = 0$ and cancels. Keeping only terms *linear* in primes:

$$\partial_t \rho' + \partial_x\big(\overline{\rho}\, u' + \rho' \overline{U}\big) + \partial_y\big(\overline{\rho}\, v'\big) \;\Rightarrow\; \partial_t \rho' + \overline{U}\partial_x \rho' + \overline{\rho}\,\partial_x u' + \partial_y(\overline{\rho} v') = 0. \tag{7}$$

Now apply the ansatz ($\partial_t \to -\mathrm{i}\omega = -\mathrm{i}\alpha c$, $\partial_x \to \mathrm{i}\alpha$, $\partial_y \to \mathrm{D}$) and substitute ρ' from Step 5; collecting terms and dividing by $\overline{\rho}$ gives exactly Eq. (8) below. *The other three equations follow by the identical procedure on* (2)–(3); the algebra is mechanical but long, so only the result is printed.

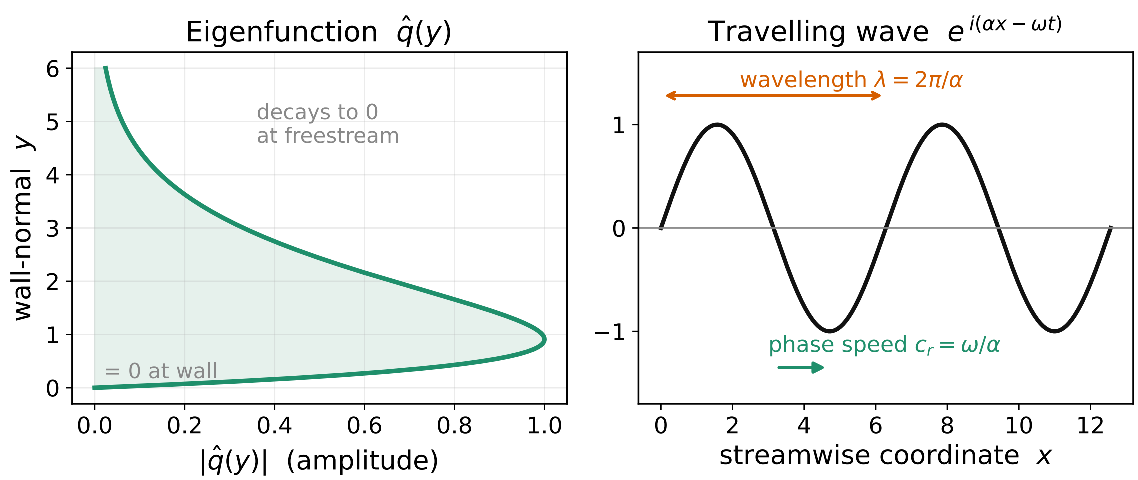

Figure 4: The ansatz (6) factorises a disturbance $q' = \hat{q}(y)\, e^{\mathrm{i}(\alpha x - \omega t)}$ into a wall-normal *shape* $\hat{q}(y)$ (left; zero at the wall, decaying at the edge — the eigenfunction) times a travelling wave in x (right; wavelength $2\pi/\alpha$, phase speed $c_r = \mathrm{Re}(c)$).

4 The four disturbance equations

What problem is solved here. These are the linear ODEs in y for $\hat{u}, \hat{v}, \hat{T}, \hat{p}$. Fixing one of α or ω real makes the other (through c or α) the eigenvalue. Every overbarred quantity is a known base-flow coefficient; all transport derivatives ($\mathrm{d}\mu/\mathrm{d}T$, $\mathrm{d}\kappa/\mathrm{d}T$, ...) are evaluated at the local mean temperature $\overline{T}(y)$ and are therefore known functions of y.

The form assembled in `pymack/temporal_solver.py` follows. Momentum and energy are divided by $\overline{\rho}$; write the viscous prefactor $\nu_R \equiv 1/(Re\,\overline{\rho})$. Throughout, every $\overline{\rho}$ is the base-flow density, i.e. the diagonal field $\overline{\rho}(y) = 1/\overline{T}(y)$ — this single substitution applies uniformly inside ν_R, $\mathcal{C}_\kappa$, and $\mathcal{C}_d$ below.

(C) Continuity + equation of state.

$$\mathrm{i}\alpha\hat{u} + \mathrm{D}\hat{v} - \frac{\overline{T}'}{\overline{T}}\hat{v} + \mathrm{i}\alpha(\overline{U} - c)\Big(\gamma M_e^2 \hat{p} - \frac{\hat{T}}{\overline{T}}\Big) = 0. \tag{8}$$

(X) Streamwise momentum.

$$\begin{aligned}\mathrm{i}\alpha(\overline{U} - c)\hat{u} + \overline{U}'\hat{v} + \mathrm{i}\alpha\overline{\rho}^{-1}\hat{p} = \nu_R\Big[&\overline{\mu}\big(\mathrm{D}^2 - \tfrac{4}{3}\alpha^2\big)\hat{u} + \overline{\mu}'\mathrm{D}\hat{u} + \mathrm{i}\alpha\big(\tfrac{1}{3}\overline{\mu}\mathrm{D} + \overline{\mu}'\big)\hat{v} \\ &+ \tfrac{\mathrm{d}\mu}{\mathrm{d}T}\big(\overline{U}'' + \overline{U}'\mathrm{D}\big)\hat{T} + \tfrac{\mathrm{d}^2\mu}{\mathrm{d}T^2}\overline{U}'\overline{T}'\hat{T}\Big].\end{aligned} \tag{9}$$

(Y) Wall-normal momentum.

$$\mathrm{i}\alpha(\overline{U} - c)\hat{v} + \overline{\rho}^{-1}\mathrm{D}\hat{p} = \nu_R\Big[\overline{\mu}\big(\tfrac{4}{3}\mathrm{D}^2 - \alpha^2\big)\hat{v} + \tfrac{4}{3}\overline{\mu}'\mathrm{D}\hat{v} + \mathrm{i}\alpha\big(\tfrac{1}{3}\overline{\mu}\mathrm{D} - \tfrac{2}{3}\overline{\mu}'\big)\hat{u} + \mathrm{i}\alpha\tfrac{\mathrm{d}\mu}{\mathrm{d}T}\overline{U}'\hat{T}\Big]. \tag{10}$$

(E) Energy (temperature).

$$\mathrm{i}\alpha(\overline{U} - c)\hat{T} + \overline{T}'\hat{v} + (\gamma - 1)\overline{\rho}^{-1}\big(\mathrm{i}\alpha\hat{u} + \mathrm{D}\hat{v}\big) = \mathcal{C}_\kappa\mathcal{L}_c\hat{T} + \mathcal{C}_d\Big[2\overline{\mu}\overline{U}'\big(\mathrm{D}\hat{u} + \mathrm{i}\alpha\hat{v}\big) + \tfrac{\mathrm{d}\mu}{\mathrm{d}T}\big(\overline{U}'\big)^2\hat{T}\Big], \tag{11}$$

$$\mathcal{L}_c = \mathrm{D}^2 - \alpha^2 + \tfrac{1}{\kappa}\tfrac{\mathrm{d}\kappa}{\mathrm{d}T}\big(2\overline{T}'\mathrm{D} + \overline{T}''\big) + \tfrac{1}{\kappa}\tfrac{\mathrm{d}^2\kappa}{\mathrm{d}T^2}\big(\overline{T}'\big)^2, \quad \mathcal{C}_\kappa = \frac{\gamma\overline{\mu}}{Pr\,Re\,\overline{\rho}}, \quad \mathcal{C}_d = \frac{\gamma(\gamma-1)M_e^2}{Re\,\overline{\rho}}. \tag{12}$$

Reading the equations.

- $\mathrm{i}\alpha(\overline{U} - c)$ is *convection by the mean relative to the wave speed*; it vanishes at the **critical layer** $\overline{U} = c_r$, where the inviscid form of the equations becomes singular — which is why that layer governs much of the instability physics.
- D, D^2 are wall-normal derivatives; $-\alpha^2$ comes from $\partial_x^2 \mapsto -\alpha^2$.
- the $\nu_R = 1/(Re\overline{\rho})$ terms are *viscous* (the $\frac{4}{3}, \frac{1}{3}$ are Stokes's bulk viscosity; the companion *Numerical Methods* note generalises this to a ratio $\lambda/\mu = 1.2$ in the spatial closure).
- the $\mathrm{d}\mu/\mathrm{d}T, \mathrm{d}\kappa/\mathrm{d}T$ terms are *temperature-dependent transport* — essential at hypersonic speeds.
- $\mathcal{C}_\kappa\mathcal{L}_c\hat{T}$ is heat conduction; $\mathcal{C}_d[\cdots]$ is viscous dissipation.

These four equations are the endpoint of the derivation and the input to the numerics. Before handing them off, we pause to say what their solutions physically are.

5 What the modes *are*: two instability mechanisms

The disturbance equations admit discrete eigenmodes; a temporal solve returns a complex phase speed such as $c = 0.930 + 0.0200\mathrm{i}$. Here is the *physics* behind the mode labels, so that such a number becomes an understood wave.

First (Tollmien–Schlichting) mode. At *low* speed the first mode is a viscous instability ($c_r \approx$ 0.3–0.6): counter-intuitively, viscosity introduces a phase lag between u' and v' near the wall, so the Reynolds stress $-\overline{\rho}\,\overline{u'v'}$ no longer averages to zero and draws energy from the mean shear. As the Mach number rises, compressibility creates a generalised inflection point (below) and the first mode becomes increasingly *inviscid/inflectional* — governed by that point, with viscosity then often *stabilising*. It is the dominant instability only up to $M_e \approx 4$.

Second (Mack) mode — a trapped acoustic wave. Above $M_e \approx 4$ a second, usually stronger, instability appears: it is essentially *a sound wave trapped between the wall and the "relative sonic line"*, reflecting in a near-wall acoustic cavity (Figure 5). This one picture explains three facts we might otherwise have to accept blindly:

- its phase speed is $c_r \approx 1 - 1/M_e$ — the *limiting* "edge-sonic" value at which the wave is sonic relative to the edge flow, which is why the spatial-solve shift target $\alpha_0 = \omega/(1 - 1/M_e)$ works; the realised c_r sits somewhat above this and varies along the branch (0.93 vs $1 - 1/6 = 0.83$ for a typical $M_e = 6$ mode), so it is a guide, not an exact value;
- it is *frequency-tuned*: like any resonator its wavelength is set by the cavity thickness, so the most-amplified wavelength is $\sim 2\delta$, giving the order-of-magnitude scaling $f \sim U_e/2\delta$ — the second-mode frequency rises as the layer thins downstream;
- its eigenfunction has a *wall-peaked pressure* — the acoustic field in the cavity.

The organising quantity: relative Mach number. Both pictures live in the **relative Mach number** $M_{\mathrm{rel}}(y) = (\overline{U} - c_r)/\overline{a}(y)$, with $\overline{a} = \sqrt{\overline{T}}/M_e$ the local sound speed. The critical layer is $M_{\mathrm{rel}} = 0$; the relative sonic line is $|M_{\mathrm{rel}}| = 1$; the second-mode acoustic cavity is the near-wall region where $|M_{\mathrm{rel}}| > 1$ (Figure 5).

Why a compressible layer is unstable at all (the generalised inflection point). Rayleigh's theorem: an inviscid *incompressible* shear flow needs a velocity inflection point ($\overline{U}'' = 0$) to be unstable. Compressibility generalises this (Lees–Lin) to the **generalised inflection point** where $\frac{d}{dy}(\overline{\rho}\,\overline{U}') = 0$ — the necessary condition for the inviscid *first* mode. Aerodynamic heating puts a hot, low-density layer near the wall (the $\overline{T}$ bulge in a compressible base flow) and pushes that generalised inflection point up into the layer — giving the robust inviscid first-mode instability that low-speed layers lack. The *second* (Mack) mode needs no inflection point: its existence condition is instead the near-wall relative-supersonic cavity ($|M_{\mathrm{rel}}| > 1$) above.

The wall-cooling reversal (the key design lever). Cooling the wall *stabilises* the first mode (it weakens the near-wall viscous mechanism) but *destabilises* the second (it thickens the relative-supersonic cavity and raises the tuned frequency). This is why cold-walled hypersonic vehicles are second-mode dominated — and why the solver carries an isothermal-wall option.

Phase vs group velocity. c_r is the *phase* speed (crest speed); energy travels at the *group* velocity $c_g = \mathrm{d}\omega_r/\mathrm{d}\alpha_r$. It is c_g that rigorously links temporal to spatial growth (relevant when the companion *Numerical Methods* note converts between them).

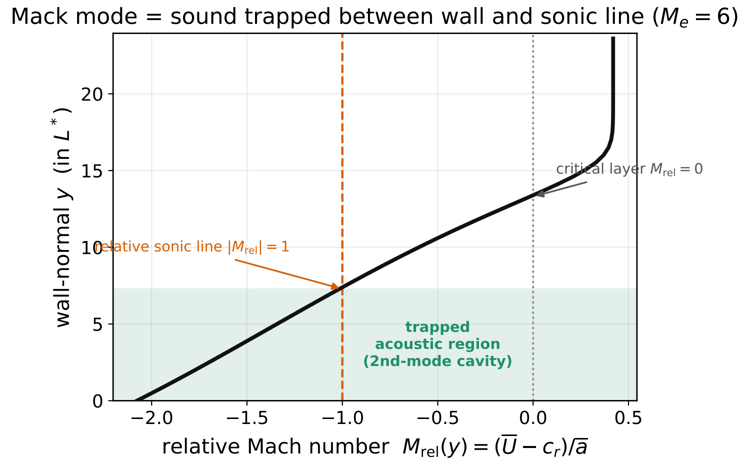

Figure 5: The second-mode mechanism, for a representative Mack mode. The relative Mach number $M_{\mathrm{rel}}(y) = (\overline{U} - c_r)/\overline{a}$ is supersonic ($|M_{\mathrm{rel}}| > 1$, shaded) in a near-wall cavity bounded by the wall and the relative sonic line ($|M_{\mathrm{rel}}| = 1$), and passes through zero at the critical layer. The Mack mode is a sound wave trapped in that cavity.

A compact comparison places any computed eigenvalue at a glance:

	First (TS) mode	**Second (Mack) mode**
mechanism	viscous (low speed) $\rightarrow$ inviscid inflectional (high speed)	trapped acoustic wave (wall–sonic line)
phase speed c_r	0.3–0.6	$\approx 1 - 1/M_e$ (here 0.93)
dominant when	low speed, $M_e \lesssim 4$	high speed, $M_e \gtrsim 4$
frequency	broadband	tuned, $f \sim U_e/2\delta$
existence criterion	generalised inflection point (Lees–Lin)	relative-supersonic cavity, $|M_{\mathrm{rel}}| > 1$
wall cooling	stabilises	destabilises
most-unstable wave	oblique (3D)	2D ($\psi = 0$)

6 Scope: the assumptions behind the equations

The equations derived above are the *2D, parallel-flow, modal* disturbance equations. That is one stage of transition (Figure 2); a careful reader should keep four idealisations in view:

1. **Parallel flow.** Setting $\overline{v} = 0$ and y-only coefficients (Step 3) neglects the $O(1/R)$ streamwise growth of the layer; the standard remedy is the *Parabolised Stability Equations* (PSE), which march the disturbance downstream retaining slow x-variation.
2. **Two-dimensionality.** The ansatz takes $\beta = 0$ (Step 4). The most-unstable *first* mode is oblique ($\psi \sim 55°$, Squire's theorem fails under compressibility), so 2D first-mode growth is a *lower bound*; the *second*/Mack mode is most unstable at $\psi = 0$, so the 2D result is its physical maximum.

3. **Modal analysis only.** Eigenvalues describe asymptotic growth; the non-normal operator also permits *transient (algebraic) growth* that modal analysis misses, relevant to bypass and blunt-body paths.
4. **Receptivity and e^N.** LST gives the amplification *ratio* $A(R)/A(R_0) = e^N$, not the initial amplitude $A(R_0)$; setting that is the separate *receptivity* problem, and e^N is an environment-dependent correlation rather than a first-principles criterion.

None of these affect the equations themselves; they bound their interpretation. A strong second (Mack) mode is captured faithfully by the 2D modal equations derived here, which is why they are the workhorse of hypersonic transition prediction. With the four ODEs (8)–(11) in hand, the next step — discretising them into matrices and solving the eigenvalue problem — is carried out in the companion *Numerical Methods* note.